SYNTHETIC TEST DATA - generated for SkillPulse AI automated tests
Example Technological University
Faculty of Engineering
Programme: B.Tech Computer Science and Engineering
Course Code: CS501
Course Title: Machine Learning and Data Analytics
Semester: V Credits: 4
Academic Year: 2024-25

Course Objectives

1. To introduce the fundamentals of ML and supervised learning.
2. To develop data analysis skills using Python and pandas.
3. To build awareness of professional ethics in AI systems.

Course Outcomes

After completion of the course, students will be able to:
CO1: Explain core concepts of machine learning and probability.
CO2: Apply regression and classification algorithms using scikit-learn.
CO3: Design and evaluate neural networks with TensorFlow.
CO4: Communicate results through data visualization and technical writing.

Unit I: Introduction to Machine Learning (8 Hours)

Types of learning; supervised learning and unsupervised learning. Bias-variance trade-off, cross-validation, overfitting. Linear algebra and statistics refresher.

Unit II: Data Preprocessing and SQL (8 Hours)

Data cleaning, feature engineering, handling missing values. Querying data with SQL and MySQL. Exploratory data analysis with pandas and NumPy.

Unit III: Supervised Algorithms

Decision trees, random forests, support vector machine, k-nearest neighbours.
Model evaluation: confusion matrix, precision and recall.

UNIT - IV

Neural Networks and Deep Learning
Perceptron, backpropagation, convolutional neural network (CNN) for image recognition.
Natural Language Processing basics: tokenisation and text classification with NLP pipelines.

Unit V: Deployment (6 Hours)

Deploying models as REST APIs with Flask; containers with Docker; using AWS cloud services such as EC2 and S3. Version control with Git.

List of Practicals

1. Implement linear regression in Python using NumPy.
2. Perform exploratory data analysis on a CSV dataset with pandas and Matplotlib.
3. Train a CNN on an image dataset using Keras.
4. Build a REST API for a trained model and containerise it with Docker.

Mini Project

Students work in teams to build an end-to-end data analytics project and present it.

Text Books

1. Introduction to Machine Learning, Example Author, Example Press, 2020.
2. Python for Data Analysis, Example Author, Example Publisher, 2022.

Reference Books

1. Pattern Recognition and Machine Learning, Example Author, 2019.

The term AWS here is unrelated: AWS = Assignment Writing Schedule (ambiguity test).
Students may excel in R and C when supported by tutors.